

Q1 JEE Main 2020 - 7 Jan (Morning)

Let the function, $f : [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for $x \in (-7, 0)$, then for all such functions f , $f(-1) + f(0)$ lies in the interval :

- (A) $[-5, -7]$
- (B) $(-\infty, 6]$
- (C) $(-\infty, 20]$
- (D) $[-5, 3]$

Q2 JEE Main 2020 - 7 Jan (Morning)

$\lim_{x \rightarrow 2} \frac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$ is equal to

Q3 JEE Main 2020 - 7 Jan (Morning)

Let S be the set of points where the function, $f(x) = |2 - |x - 3||$, $x \in \mathbb{R}$ is not differentiable. Then $\sum_{X \in S} f(f(x))$ is equal to

Q4 JEE Main 2020 - 7 Jan (Evening)

Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If

$\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3} \right) = 4$, then which one of the following is not true?

- (A) $f(x)$ has minima at $x = 1$ & maxima at $x = -1$
- (B) $f(1) - 4f(-1) = 4$
- (C) $f(x)$ is maxima at $x = 1$ and minima at $x = -1$
- (D) $f(x)$ is odd

Q5 JEE Main 2020 - 7 Jan (Evening)

The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is :

- (A) $\frac{4-\sqrt{7}}{3}$
- (B) $\frac{4+\sqrt{5}}{3}$
- (C) $\frac{4+\sqrt{7}}{3}$
- (D) $\frac{4+\sqrt{5}}{3}$

Q6 JEE Main 2020 - 7 Jan (Evening)

If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+3x}{1-2x} \right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$ is

continuous, then k is equal to

Q7 JEE Main 2020 - 8 Jan (Morning)

$\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is equal to

- (A) e^{-2}
- (B) e^2
- (C) $e^{2/7}$
- (D) $e^{3/7}$

Q8 JEE Main 2020 - 8 Jan (Evening)

$\lim_{x \rightarrow 0} \frac{\int_0^x t \sin(10t) dt}{x}$ is equal to

- (A) 1
- (B) 10
- (C) 5
- (D) 0

Q9 JEE Main 2020 - 9 Jan (Morning)

$f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} & ; x < 0 \\ b & ; x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}} & ; x > 0 \end{cases}$

Function is continuous at $x = 0$, find $a + 2b$.

- (A) 0
- (B) 4
- (C) 6
- (D) 8

Q10 JEE Main 2020 - 9 Jan (Morning)

If $f(x)$ is twice differentiable and continuous function in $x \in [a, b]$ also $f'(x) > 0$ and $f''(x) < 0$ and $c \in (a, b)$ then $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than

(A) $\frac{b-c}{c-a}$

(B) 1

(C) $\frac{a+b}{b-c}$

(D) $\frac{c-a}{b-c}$

Q11 JEE Main 2020 - 9 Jan (Evening)

Let $[t]$ denote the greatest integer $\leq t$ and $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$. Then the function, $f(x) = [x^2] \sin(\pi x)$ is discontinuous, when x is equal to :

(A) $\sqrt{A}$

(B) $\sqrt{A+1}$

(C) $\sqrt{A+5}$

(D) $\sqrt{A+21}$

Q12 JEE Main 2020 - 2 Sep (Morning)

If a function $f(x)$ defined by

$$f(x) = \begin{cases} ae^x + be^{-x}, & -1 \leq x < 1 \\ cx^2, & 1 \leq x \leq 3 \\ ax^2 + 2cx, & 3 < x \leq 4 \end{cases}$$

be continuous for some $a, b, c \in R$ and $f'(0) + f'(2) = e$, then the value of a is :

(A) $\frac{e}{e^2+3e+13}$

(B) $\frac{e}{e^2-3e+13}$

(C) $\frac{e}{e^2-3e-13}$

(D) $\frac{1}{e^2-3e+13}$

Q13 JEE Main 2020 - 2 Sep (Morning)

If $\lim_{x \rightarrow 1} \frac{x+x^2+x^3+\dots+x^n-n}{x-1} = 820$, ($n \in N$) then the value of n is equal to

Q14 JEE Main 2020 - 2 Sep (Evening)

$\lim_{x \rightarrow 0} \left(\tan\left(\frac{\pi}{4} + x\right) \right)^{\frac{1}{x}}$ is equal to :

- (A) e^2
 (B) 1
 (C) e
 (D) 2

Q15 JEE Main 2020 - 3 Sep (Morning)

Let $[t]$ denote the greatest integer $\leq t$. If for some $\lambda \in \mathbf{R} - \{0, 1\}$, $\lim_{x \rightarrow 0} \left| \frac{1-x+|x|}{\lambda-x+[x]} \right| = L$, then L is equal to :

- (A) 2
 (B) $\frac{1}{2}$
 (C) 0
 (D) 1

Q16 JEE Main 2020 - 3 Sep (Morning)

If $\lim_{x \rightarrow 0} \left\{ \frac{1}{x^8} \left(1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right) \right\} = 2^{-k}$, then the value of k is

Q17 JEE Main 2020 - 3 Sep (Evening)

$\lim_{x \rightarrow a} \frac{(a+2x)^{\frac{1}{3}} - (3x)^{\frac{1}{3}}}{(3a+x)^{\frac{1}{3}} - (4x)^{\frac{1}{3}}} (a \neq 0)$ is equal to

- (A) $\left(\frac{2}{9}\right) \left(\frac{2}{3}\right)^{\frac{1}{3}}$
 (B) $\left(\frac{2}{3}\right) \left(\frac{2}{9}\right)^{\frac{1}{3}}$
 (C) $\left(\frac{2}{3}\right)^{\frac{4}{3}}$
 (D) $\left(\frac{2}{9}\right)^{\frac{4}{3}}$

Q18 JEE Main 2020 - 4 Sep (Morning)

Suppose a differentiable function $f(x)$ satisfies the identity

$f(x+y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3)$ is

equal to

Q19 JEE Main 2020 - 4 Sep (Evening)

Let $f : (0, \infty) \rightarrow (0, \infty)$ be a differentiable function such that $f(1) = e$ and

$$\lim_{t \rightarrow x} \frac{t^2 f^2(x) - x^2 f^2(t)}{t - x} = 0 \text{ If } f(x) = 1, \text{ then } x \text{ is equal to :}$$

- (A) $2e$
- (B) e
- (C) $\frac{1}{2e}$
- (D) $\frac{1}{e}$

Q20 JEE Main 2020 - 4 Sep (Evening)

The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$ is :

- (A) both continuous and differentiable on $\mathbb{R} - \{-1\}$
- (B) both continuous and differentiable on $\mathbb{R} - \{1\}$
- (C) continuous on $\mathbb{R} - \{-1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$
- (D) continuous on $\mathbb{R} - \{1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$

Q21 JEE Main 2020 - 5 Sep (Morning)

If the function $f(x) = \begin{cases} k_1(x - \pi)^2 - 1, & x \leq \pi \\ k_2 \cos x, & x > \pi \end{cases}$ is twice differentiable, then the ordered pair

(k_1, k_2) is equal to:

- (A) $(1, 0)$
- (B) $(\frac{1}{2}, 1)$
- (C) $(1, 1)$
- (D) $(\frac{1}{2}, -1)$

Q22 JEE Main 2020 - 5 Sep (Morning)

If α is the positive root of the equation, $p(x) = x^2 - x - 2 = 0$, then $\lim_{x \rightarrow \alpha^+} \frac{\sqrt{1 - \cos(p(x))}}{x + \alpha - 4}$ is equal to :

- (A) $\frac{1}{\sqrt{2}}$
- (B) $\frac{1}{2}$

(C) $\frac{3}{\sqrt{2}}$

(D) $\frac{3}{2}$

Q23 JEE Main 2020 - 5 Sep (Morning)

Let $f(x) = x \cdot \left[\frac{x}{2} \right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to

Q24 JEE Main 2020 - 5 Sep (Evening)

$$\lim_{x \rightarrow 0} \frac{x \left(e^{(\sqrt{1+x^2+x^4}-1)/x-1} \right)}{\sqrt{1+x^2+x^4}-1}$$

(A) is equal to 0.

(B) is equal to $\sqrt{e}$.

(C) is equal to 1.

(D) does not exist.

Q25 JEE Main 2020 - 6 Sep (Morning)

$$\lim_{x \rightarrow 1} \left(\frac{\int_0^{(x-1)^2} t \cos(t^2) dt}{(x-1) \sin(x-1)} \right)$$

(A) does not exist

(B) is equal to $-\frac{1}{2}$

(C) is equal to 1

(D) is equal to $\frac{1}{2}$

Q26 JEE Main 2020 - 6 Sep (Evening)

Let $f : R \rightarrow R$ be a function defined by $f(x) = \max \{x, x^2\}$. Let S denote the set of all points in R , where f is not differentiable. Then :

(A) ϕ (an empty set)

(B) $\{0, 1\}$

(C) $\{1\}$

(D) $\{0\}$

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Answer Key

Q1 (C)	Q2 (36)	Q3 (3)	Q4 (A)
Q5 (A)	Q6 (5)	Q7 (A)	Q8 (D)
Q9 (A)	Q10 (D)	Q11 (B)	Q12 (B)
Q13 (40)	Q14 (A)	Q15 (A)	Q16 (8)
Q17 (B)	Q18 (10)	Q19 (D)	Q20 (D)
Q21 (B)	Q22 (C)	Q23 (8)	Q24 (C)
Q25 No option is correct	Q26 (B)		